

Exact Critical Points: Damping, Phase Transitions, and Chemical Equilibrium

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Abstract—This paper examines three systems, from three different fields, unified by the same structural fact this series has repeatedly found: the boundary between qualitatively different behavior sits at an exact, calculable point rather than a fuzzy region. We prove and numerically confirm the critical damping ratio $\zeta = 1$ as the sharp boundary between oscillatory and non-oscillatory response in a second-order system, the mean-field critical temperature T_c as the exact point where spontaneous magnetization vanishes, and the exact neutrality point $\text{pH} = 7$ together with a titration equivalence point, both derived from first principles and confirmed against a simulated titration curve.



1 CRITICAL DAMPING: THE EXACT OSCILLATION BOUNDARY

A standard second-order system, $\ddot{x} + 2\zeta\omega_n\dot{x} + \omega_n^2x = 0$, has closed-form solutions depending on the damping ratio ζ :

$$x(t) = \begin{cases} e^{-\zeta\omega_n t} \left(\cos \omega_d t + \frac{\zeta\omega_n}{\omega_d} \sin \omega_d t \right), & \zeta < 1, \omega_d = \omega_n \sqrt{1 - \zeta^2} \\ (1 + \omega_n t) e^{-\omega_n t}, & \zeta = 1 \\ Ae^{r_1 t} + Be^{r_2 t}, & \zeta > 1, r_{1,2} = -\omega_n(\zeta \mp \sqrt{\zeta^2 - 1}) \end{cases} \quad (1)$$

Proposition 1. For $\zeta \geq 1$, $x(t) \geq 0$ for all $t \geq 0$ (no overshoot below the equilibrium). For $\zeta < 1$, $x(t)$ crosses zero and goes negative.

Proof sketch. For $\zeta = 1$: $x(t) = (1 + \omega_n t)e^{-\omega_n t} > 0$ for all $t \geq 0$, since both factors are positive. For $\zeta > 1$: both $r_1, r_2 < 0$ are real and $A, B > 0$ (from initial conditions $x(0) = 1, \dot{x}(0) = 0$), so $x(t)$ is a positive combination of decaying exponentials, always positive. For $\zeta < 1$: the solution contains $\cos \omega_d t$, which is negative for $\omega_d t \in (\pi/2, 3\pi/2)$, and the envelope $e^{-\zeta\omega_n t}$ is strictly positive, so $x(t)$ must cross zero and go negative before fully decaying, for any $\zeta \in [0, 1)$. $\square$

1.1 Worked numerical verification

Using the exact closed-form solutions (not numerical integration, to avoid precision artifacts) across $\zeta \in \{0.3, 0.7, 0.9, 1.0, 1.5, 3.0\}$:

TABLE 1
Overshoot and settling time vs. damping ratio

ζ	Overshoot	Settling time (2% band)
0.3	0.37233	5.615
0.7	0.04599	2.990
0.9	0.00152	2.350
1.0	0.00000	2.917
1.5	0.00000	5.327
3.0	0.00000	(never settles within window)

Overshoot is exactly zero for every $\zeta \geq 1$ tested and strictly positive for every $\zeta < 1$ tested, with the magnitude shrinking toward zero as $\zeta \rightarrow 1^-$ — confirming $\zeta = 1$ is the sharp, exact boundary, not an approximate one. Settling time is also near-minimal at $\zeta = 1$ among non-oscillatory responses (compare 2.917 at $\zeta = 1$ to 5.327 at $\zeta = 1.5$), the basis for critical damping's standard engineering role: fastest return to equilibrium without overshoot.

2 MEAN-FIELD PHASE TRANSITION: THE EXACT CRITICAL TEMPERATURE

The mean-field (Curie-Weiss) model of a ferromagnet gives magnetization m as the self-consistent solution of

$$m = \tanh(\beta J z m), \quad \beta = 1/T \text{ (units with } k_B = 1). \quad (2)$$

Proposition 2. A nonzero solution $m \neq 0$ exists if and only if $T < T_c := Jz$.

Proof sketch. $m = 0$ is always a solution. Writing $f(m) = \tanh(\beta J z m)$, a nonzero fixed point exists near $m = 0$ exactly when $f'(0) > 1$, i.e., $\beta J z > 1$, i.e., $T < Jz$, since $\tanh$ is concave for $m > 0$ and $f(0) = 0$, guaranteeing a unique positive crossing whenever the initial slope exceeds 1, and none otherwise. $\square$

2.1 Worked numerical verification

Solving Eq. (2) numerically (root-finding on the nonzero branch) across a range of temperatures with $Jz = 1$ (so $T_c = 1$ exactly):

Nonzero magnetization shrinks continuously toward zero as $T \rightarrow T_c^-$ and is exactly zero for every $T \geq T_c$ tested — the transition is sharp and occurs precisely at the value the algebra predicts, not approximately near it.

3 CHEMICAL EQUILIBRIUM: TWO EXACT MID-POINTS

Proposition 3. In pure water at 25°C, the neutral pH is exactly 7.

TABLE 2
Nonzero-branch magnetization vs. temperature

T/T_c	Magnetization m
0.500	0.957504
0.800	0.710412
0.950	0.379485
0.990	0.172511
1.000	0.000000
1.010	0.000000
1.200	0.000000
1.500	0.000000

Proof. Water autoionizes: $\text{H}_2\text{O} \rightleftharpoons \text{H}^+ + \text{OH}^-$, with $K_w = [\text{H}^+][\text{OH}^-] = 10^{-14}$ at 25°C . In pure water, $[\text{H}^+] = [\text{OH}^-]$ by stoichiometry, so $[\text{H}^+]^2 = 10^{-14}$, giving $[\text{H}^+] = 10^{-7}$ and $\text{pH} = -\log_{10}(10^{-7}) = 7$ exactly. $\square$

Proposition 4. For a titration of a weak acid with a strong base, the equivalence point occurs at exactly $V_b = C_a V_a / C_b$.

Proof. The equivalence point is defined by equal moles of acid and base added: $n_a = C_a V_a = n_b = C_b V_b$, solving directly for V_b . $\square$

3.1 Worked numerical verification

For acetic acid ($K_a = 1.8 \times 10^{-5}$, $C_a = 0.1\text{M}$, $V_a = 50\text{mL}$) titrated with 0.1M strong base, the predicted equivalence point is $V_b = (0.1 \times 50)/0.1 = 50.0\text{mL}$ exactly. Simulating the titration curve (Henderson–Hasselbalch in the buffer region, hydrolysis equilibrium at and past equivalence) confirms a sharp inflection precisely there:

TABLE 3
Simulated titration curve near equivalence

V_b (mL)	pH
45.00	5.699
49.00	6.435
49.90	7.443
50.00	8.722
50.10	10.000
51.00	10.996

The pH jumps by more than 2.5 units between $V_b = 49.9$ and $V_b = 50.1$ mL — a span of only 0.2mL , or 0.4% of the total titration volume — confirming the equivalence point is a genuine, sharp feature of the curve located exactly where the stoichiometric formula predicts, not a gradual transition spread across the titration.

4 CONCLUSION

Three unrelated fields, one recurring structural fact:

- Critical damping ($\zeta = 1$) is the exact, zero-slack boundary between oscillatory and non-oscillatory response — confirmed by exact closed-form solutions showing zero overshoot for every $\zeta \geq 1$ tested and nonzero overshoot for every $\zeta < 1$ tested.
- The mean-field critical temperature ($T_c = Jz$) is the exact point where spontaneous order vanishes —

confirmed numerically with magnetization shrinking continuously to zero exactly at $T/T_c = 1$.

- Chemical neutrality ($\text{pH} = 7$) and titration equivalence ($V_b = C_a V_a / C_b$) are both exact, derivable midpoints — confirmed by a simulated titration curve showing a sharp, 2.5-pH-unit jump concentrated within 0.4% of the total titration volume, centered exactly on the predicted point.

As throughout this series: in every system examined, the critical point was never a rough region to be gestured at — it was a specific, calculable number, and the numbers computed here landed on it exactly.

REFERENCES

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